

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Vantage Santa Clara CA2, CA -- station KSJC, America/Los_Angeles
Committed pair OSM 1314010354 -> 1075445245
Vantage Santa Clara CA2 / OSM way 1075445245
Facade gap 350.5 m between the two halls
Weather record 43,747 real hourly records from KSJC
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.2056 C at 305 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2021-01-03 (clear_cool)
Plant limit 18.0 C
Notice required 6 h
Forecast skill 0.00 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 6 of 24 hours with 2 mode changes. The reactive incumbent took 10 hours -- 4 more -- but declared 0 unsafe hours against the agent's 0. 10 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 6 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent 10 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
10 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	-	9.050	18.0	10.560	3.098
01	mechanical	yes	-	9.600	18.0	10.560	2.663
02	mechanical	yes	-	10.160	18.0	10.560	2.525
03	mechanical	no	refusal	10.239	18.0	10.560	2.063
04	FREE	yes	-	10.710	18.0	10.560	2.354
05	FREE	yes	-	10.710	18.0	10.560	2.044
06	FREE	yes	-	11.260	18.0	10.560	2.012
07	FREE	yes	-	12.295	18.0	11.110	1.818
08	FREE	yes	-	14.600	18.0	11.110	2.204
09	FREE	yes	-	17.370	18.0	11.110	2.891
10	mechanical	no	dry-bulb	20.150	18.0	12.220	3.656
11	mechanical	no	refusal	22.904	18.0	13.330	4.602

12	mechanical	no	refusal	23.991	18.0	12.780	4.614
13	mechanical	no	dry-bulb	24.590	18.0	13.890	4.779
14	mechanical	no	refusal	22.350	18.0	13.498	3.942
15	mechanical	no	refusal	20.124	18.0	12.780	3.790
16	mechanical	yes	-	17.861	18.0	12.780	3.119
17	mechanical	yes	-	15.700	18.0	12.220	2.719
18	mechanical	yes	-	12.370	18.0	12.220	2.505
19	mechanical	no	refusal	11.761	18.0	11.670	2.880
20	mechanical	yes	switch budget	10.604	18.0	11.670	3.498
21	mechanical	yes	switch budget	9.600	18.0	10.560	3.457
22	mechanical	yes	switch budget	10.093	18.0	11.110	3.568
23	mechanical	yes	switch budget	10.700	18.0	11.110	4.002

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.050 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.600 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.160 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.710 C, which is 7.290 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.354 C, and the margin is measured, not chosen: 2.204 C of group-conditional forecast error for this hour of day, plus 0.1504 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.710 C, which is 7.290 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.044 C, and the margin is measured, not chosen: 1.893 C of group-conditional forecast error for this hour of day, plus 0.1504 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.260 C, which is 6.740 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.012 C, and the margin is measured, not chosen: 1.862 C of group-conditional forecast error for this hour of day, plus 0.1504 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.295 C, which is 5.705 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.818 C, and the margin is measured, not chosen: 1.744 C of group-conditional forecast error for this hour of day, plus 0.0747 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.600 C, which is 3.400 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.204 C, and the margin is measured, not chosen: 2.054 C of group-conditional forecast error for this hour of day, plus 0.1504 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.370 C, which is 0.630 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.891 C, and the margin is measured, not chosen: 2.741 C of group-conditional forecast error for this hour of day, plus 0.1504 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.150 C against a 18.0 C limit, so it fails by 2.150 C. It would take a limit 2.150 C higher, or a bound 2.150 C tighter, to change this hour.

11:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

12:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.590 C against a 18.0 C limit, so it fails by 6.590 C. It would take a limit 6.590 C higher, or a bound 6.590 C tighter, to change this hour.

14:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

15:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

16:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.861 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.700 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.370 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.604 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.600 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.093 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.700 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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